

EEHC DISTRIBUTION MATERIALS SPECIFICATION	EDMS 21 - 301 - 1
shunt power capacitors of the self-healing type for low voltage network	11-02-2025

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SPECIFICATION

FOR

SHUNT POWER CAPACITORS OF THE SELF-HEALING TYPE FOR low voltage network

UP TO 50 KVAR

Issue: February. 2025/ Rev-1

- This revision contains option items that must be selected by EDC before bidding.
- **Glossary of Common Abbreviations**
 - **EDMS** – Electricity Distribution Materials Specification
 - **EDC** – Egyptian Electricity Company
 - **PSD** – Pressure Sensitive Disconnection
 - **PFC** – Power Factor Correction
 - **MPP** – Metallized Polypropylene
 - **THD** – Total Harmonic Distortion
 - **RMS** – Root Mean Square
 - **VAR** – Volt-Ampere Reactive
 - **(GWFI)** - Glow-Wire Flammability Index

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1. SCOPE:

This specification specifies the minimum technical requirements for design, manufacturing, testing, inspection, supply and delivery **of low voltage capacitors**. Such equipment shall be utilized for the improvement of power factor and/or mitigation of harmonics in AC electrical power systems; this is in addition to special requirement of Distribution Companies (EDCs).

2. APPLICABLE STANDARDS:

The equipment/material covered in this specification shall comply with the latest edition/amendment of the standards given in Table (1). Where any provision of this specification differs from those of the standards listed below, the provisions of this specification shall apply. Any deviations from the listed standards or the provisions of this specification should be clearly set out in the offer.

Table (1)

Standard No.	Description
IEC60831 Shunt power capacitors	Part 1 General – Performance, testing and rating – Safety requirements – Guide for installation and operation
	Part 2 Ageing test, self-healing test and destruction test
IEC 61921	Power capacitors-low –voltage power factor correction banks
IEC 61557-1	Electrical safety in low voltage distribution systems up to 1000VAC
IEC 6 0695-2-12	Fire hazard testing – Part 2-12: Glowing/hot-wire based test methods – Glow-wire flammability index (GWFI) test method for materials
IEEE 18	Standard for Shunt Capacitor
IEEE 519	Application of Harmonic Limits

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3. ENVIRONMENTAL CONDITIONS

The electrical and mechanical properties of the required low voltage capacitors should be guaranteed under the environmental conditions given in Table (2).

Table (2)

Ambient temperature	-5°C to +55°C (Class D)
Maximum relative humidity	95 %
Altitude	Up to 1000 m above sea-level

4. DESIGN CRITERIA

4.1 Capacitor type

- The capacitor unit only shall be dry type (gas –filled or resin material without any leakage) according to EDC requirements.
- The thickness of the dielectric depends on the voltage rating. Metallization and edge enhancement, including additional junctions or cross-profile metallization, play a crucial role in achieving high current handling capability and ensuring stable capacitance at elevated operating temperatures.

4.2 Construction

Casing	Extruded Aluminium cylindrical
Dielectric	Metallized polypropylene film with Zn/Al alloy

4.3 Self-healing

- The capacitors shall have self-healing ability, where damage to a part of the dielectric, caused by a microscopic flaw in the dielectric film due to over-voltage, can be quickly self-healed, restoring the capacitor to its normal state. This significantly enhances reliability.
- The capacitor's features, including a self-healing MPP dielectric impregnated with PU resin, promote effective stress management, low temperature rise, and improved harmonic current handling. With a special MPP film, the capacitor can continuously handle up

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to **1.43** times its rated current and ensures safe disconnection at the end of its lifespan

4.4 (PSD) mechanism.

The power capacitors has a three-phase design with a peripheral Pressure Sensitive Disconnection (PSD) mechanism.

4.5 Capacitance tolerances

The capacitance shall not differ from the rated capacitance by more than

- **-5 %** to **+10 %** for units and banks

In three-phase units, the ratio of maximum to minimum value of the capacitance measured between any two-line terminals shall not exceed **1.08**.

4.6 Losses in steady state rated Voltage and rated frequency At 25°C indoor

- Dielectric losses : Not more than 0.2 W/KVAR
- Total losses : Not more than 0.5 W/KVAR

4.7 Self-Extinguishing

The insulation material used in the capacitors must be **Self-Extinguishing Resin** with a **V0 rating**(Self-extinguishing materials with a minimum Glow-Wire Flammability Index (GWFI) of 750 °C shall be used for the terminals) according to the **UL 94** or **IEC 60695-2-12** standards, featuring the following properties:

1. Capability of self-extinguishing within 10 seconds after the flame source is removed.
2. No production of flaming droplets during ignition.
3. High level of safety against fire hazards, ensuring compliance with electrical safety requirements.

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5. CONSTRUCTION REQUIREMENTS

5.1 General

- The power capacitors shall be operable on 50 Hz, **400** volt and shall meet or comply with industry standards.
- The rated admissible voltage values **440/480/525 V** (in accordance with EDC requirements).
- Power capacitors shall be installed in cool, well-ventilated locations, away from other heat-radiating elements. The capacitors shall be non-polluting, non-toxic, and environmentally friendly.
- Except for capacitors directly connected to motor terminals, all capacitor banks shall be equipped with a switching device featuring an automatic trip function
- **Over voltage As per IEC 60831 Part 1 and 2**

Admissible voltage levels in service	
VOLTAGE FACTOR XU_N R.M.S	DURATION
100% U_N	CONTINUOUS
110% U_N	8 hours per day
115% U_N	30 minutes per day
120% U_N	5 minutes
130% U_N	1 minute

- Over current **1.43** x I_N with harmonics, or more than as per EDC request
- Normal switching frequency: not less than 5,000 operations per year at rated ambient temperature (class D).
- Mean life expectancy: not less than 130,000 hours at rated ambient temperature (class D).
- **Discharge device:** Each capacitor unit shall be provided with a means for

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discharging (resistance) each unit in 3 min to 75 V or less .

- Power capacitors shall be capable of being mounted in both vertical and horizontal positions.
- The protection rating of the capacitors shall be no less than IP20 and IP 54 as per EDC request.
- Additionally, adequate protection shall be provided against dust and direct solar radiation.
- Fastening: Threaded M10 or M12 stud at the bottom.

5.2 Inrush current

- PFC capacitors shall be capable of withstanding high inrush currents caused by switching operations. For capacitors connected in parallel (i.e., in banks), an inrush current of up to 200 times the rated current may occur.
- An inrush current exceeding 200 times the rated current shall be optional, based on EDC requirements.

6. SAFETY REQUIREMENTS

- Power capacitors shall be installed in a suitable ventilated location and not close to objects that dissipate heat, like filter circuit reactors and furnaces or in direct sunlight.
- The power capacitors shall contain non-accessible, built-in discharge resistors to ensure user safety by avoiding user contact with live terminals.

7. TESTING and INSPECTION

7.1 General

- Tests should be performed at the manufacturer's work in accordance with the relevant latest IEC standards.

7.2 Type Tests:

The following tests are type tests. For details, reference should be made to the relevant clauses or subclauses:

- a) thermal stability test;

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- b) measurement of the tangent of the loss angle ($\tan \delta$) of the capacitor at elevated temperature ;
- c) voltage test between terminals ;
- d) voltage test between terminals and container ;
- e) lightning impulse voltage test between terminals and container ;

The impulse test shall be performed with a wave of 1,2/50 μ s to 5/50 μ s having a peak value of **8 kV** if the rated voltage of the capacitor is $U_N \leq 690$ V

- f) discharge test ;
 - g) ageing test ;
 - h) self-healing test;
 - i) destruction test.
- Type tests are carried out in order to ascertain that, as regards design, size, materials and construction, the capacitor complies with the specified characteristics and operation requirements detailed in this standard.
 - Unless otherwise specified, every capacitor sample to which the type test is applied shall first have withstood satisfactorily the application of all the routine tests
 - The type tests shall have been carried out by the manufacturer, and the purchaser shall, on request, be supplied with a certificate detailing the results of such tests.
 - The successful completion of each type test is also valid for units

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having the same rated voltage and lower output, provided that they do not differ in any way that may influence the properties to be checked by the test. It is not essential that all type tests be carried out on the same capacitor sample.

- The number of samples for the type test shall be subjected to agreement between the manufacturer and user.

7.3 Routine Tests:

The following tests are routine tests. For details, reference should be made to the relevant clauses or subclauses:

- Inspection for conformity with specifications.
- capacitance measurement and output calculation ;
- measurement of the tangent of the loss angle ($\tan \delta$) of the capacitor ;
- voltage test between terminals;**
 - Each capacitor shall be subjected to an a.c. test at $U_t = 2,15 U_N$ for a minimum time of 2 s.
- voltage test between terminals and container ;**
 - The voltage to be applied is $2 U_N + 2 \text{ kV}$ or **3 kV**, whichever is the higher, for **10 s** or **20 %** higher for a minimum time of 2 s.
- test of the internal discharge device ;
- sealing test .

Routine tests shall have been carried out by the manufacturer on every capacitor before delivery. If the purchaser so requests, he

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shall be supplied with a certificate detailing the results of such tests.

In general, the indicated sequence of the tests is not mandatory.

8. SUBMITTALS

- The supplier should provide with shipment of the capacitor complete installation, operation and maintenance instruction/manuals suitable for the specific supplied item(s), including information from the manufacturer of each purchased accessory.
- All manuals, instructions and supporting drawings and parts lists supplied by the manufacturer and supplemental instruction and data supplied by accessory manufacturers should be in English, clearly readable and arranged in such format to make the subject items easily identifiable.
- The supplier should provide a complete set of “as manufactured” drawings and wiring diagrams including interconnection diagrams of all components and assemblies of the equipment which assist in operation and maintenance of the equipment.
- All drawings should be compatible with current industrial standards and good industrial practice.
- All information should be in English language, clearly readable, with IEC standard symbols, device model, numbers and other requirements.

9. GUARANTEE

- The supplier guarantee the capacitors against all defects arising out of faulty design or workmanship, or of defective material for a period of 12 months from putting in service or 18 months from delivery date.

10. Marking of the unit :

10.1 Rating plate

- The following information shall be marked indelibly, either directly or by means of a plate, on each capacitor unit:
 - a) Manufacturer.

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- b) Identification number and manufacturing year.(The year may be a part of the identification number or be in code form)
- c) Rated output **QN** in (**KVAR**).
- d) Rated voltage **UN** in volts (**V**).
- e) Rated frequency **FN** in hertz (**Hz**).
- f) Temperature category.
- g) Discharge device, if internal, shall be indicated by wording or by the symbol  or by the rated resistance in kilohms (**kΩ**) or megohms (**MΩ**).
- h) Reference of self-healing design: "SH" or  or "self-healing".
- i) Connection symbol. (All capacitors, except single-phase units having one capacitance only, shall have their connection indicated. For standardized connection symbols).
- j) Internal fuses, if included, shall be indicated by wording or by the symbol  .
- k) Indication for the overpressure or thermal disconnecter, if such disconnecter is fitted.
- l) Insulation level **Ui** in kilovolts (**kV**). (Only for units having all terminals insulated from the container).The insulation level shall be marked by means of two numbers separated by a stroke, the first number giving the r.m.s. value of the power frequency test voltage, in kilovolts, and the second number giving the peak value of the lightning impulse test voltage, in kilovolts (for example 3/15 kV).For units having one terminal permanently connected to the container, and not tested according to Clause 15, this information should be 3/- kV.
- m)Reference to IEC 6 0831-1 (plus year of issue of the edition).

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10.2 Standardized connection symbols

The type of connection shall be indicated either by letters or by the following symbols:

D or  = delta

Y or  = star

YN or  = star, neutral brought out

III or  = three sections without interconnections.

10.3 Warning plate

When capacitors are impregnated with products that shall not be dispersed into the environment

(see Clause 24), the capacitor shall carry markings in accordance with the laws or regulations in force in the user's country.

11. TECHNICAL DATA SCHEDULE

8. The tenderer must fill in thoroughly the attached technical data schedule.
9. Any offer does not accompanied with clear and complete technical data schedule shall be rejected.

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TECHNICAL DATA SCHEDULE

I/we guarantee the correctness of above data for shunt power capacitors we are offering to Electricity Distribution Company.

clauses			Specs values	Vendors values
Manufacturer's name				
Country of origin				
Camera Model				
Standards:			IEC 60831-1/2	
Voltage range:			400 V to 525V in Three Phase	
Frequency:			50 HZ	
Power range:			Up to 50 KVAR	
Losses (dielectric):			< 0.2W/KVAR	
Losses (total):			< 0.5W/KVAR	
Capacitance tolerance:			-5%, +10%	
Voltage test:	Between terminals:	2.15 x U _N (AC), 10s		
	Between terminal & container:	3 kV (AC), 10s		
	Impulse voltage:	8 kV		
<u>Working Conditions</u>				
Ambient temperature			-5°C to +55°C (class D)	
Humidity:			95%	
Altitude:			1,000 m above sea level	
Overvoltage:			1.1 x U _N , 8 h in every 24 H	
Overcurrent:			Up to 1.43 x I _N	
Peak inrush current:			200 x I _N	
Switching operations (max.):			Not less than 5,000 switching operations per year	
- Mean Life expectancy:			Not less than 130,000 H	
<u>Installation Characteristics</u>				
Mounting position:			Indoor, vertical, horizontal	

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Fastening:	Threaded M10 or M12 at the bottom	
Safety:	Self-healing + Pressure-sensitive disconnecter + Discharge device	
Protection:	IP20	
<u>Construction</u>		
Casing:	Extruded Aluminium cylindrical	
Dielectric:	Metallized polypropylene film with Zn/Al alloy or dry resin	

Signature of tenderer: Date: / /20